H3 — ML + Embedded Track

Member: ______

Your role

You own the export and sampling explainers. You curate the domain dataset for the improvise phase.

Weekly tasks

Week 1 — Export explainer

- Task: Read research/tinystories/sample.py and export.py. Write the explainer: the PLE model.bin format, golden.txt, golden.npz, tokenizer hashing, the tied head.
- Deliverable: The 1-page export explainer.
- Verify: You can explain why `--allow-unverified-tokenizer` exists.
- Learn: Neural network basics. Watch the 3Blue1Brown series on neural networks.

Week 2 — Sample checks

- Task: As runs finish, run sample.py on each arm. Record 5 sample texts per arm.
- Deliverable: The sample log with timestamps.
- Verify: Every sample decodes to readable text.
- Learn: Embeddings and next-token prediction. Explain what the model predicts at each step.

Week 3 — 20 samples

- Task: Generate 20 samples from the trained models. Write coherence notes per arm.
- Deliverable: The sampling notes table.
- Verify: The ple samples are more coherent than the baseline samples, as published.
- Learn: The transformer block. Explain the generation loop.

Week 4 — Code walkthrough

- Task: Write the code walkthrough of the runtime: attention, output head, PLE path.
- Deliverable: RUNTIME_WALKTHROUGH.md in the club repo.
- Verify: E1 reads it and finds no factual error.
- Learn: Sampling and generation. Explain greedy vs sampled decoding.

Week 5 — Domain dataset

- Task: Curate a small domain dataset for the improvise direction: a few thousand examples. Tokenize with the repo tokenizer.
- Deliverable: The domain dataset in the club Drive with a README.
- Verify: The data passes the tokenizer without errors.
- Learn: RoPE and KV cache. Draw the cache growth over 200 tokens.

Week 6 — Domain quality checks

- Task: Generate 20 domain samples from the fine-tuned model. Score coherence.
- Deliverable: The domain quality table.
- Verify: The scores beat the untuned model on the same prompts.
- Learn: Fine-tuning. Explain why a small dataset works for a small model.

Week 7 — Teach tokenization

- Task: Present tokenization and embeddings to the team at the meetup.
- Deliverable: Your teach-back slides.
- Verify: R3 can explain why vocab 32768 costs memory.
- Learn: Prepare your presentation deeply.

Week 8 — Repo cleanup

- Task: Clean the club repo: docs, runbook, artifacts, README.
- Deliverable: A clean repo that a new member can clone and run.
- Verify: Follow your own README from a fresh clone.
- Learn: Write the "what I learned" page.

Learning path

1. Neural network basics: perceptron, MLP, gradient descent.
2. Embeddings, tokenization, next-token prediction.
3. Transformer block: attention, LayerNorm, residuals, MLP.
4. Loss, perplexity, sampling, generation.
5. Attention math, RoPE, KV cache. PLE paper.
6. Fine-tuning, transfer learning, PTQ.
7. Teach one topic to the team.
8. Write the "what I learned" page.

Resources

- Repo: https://github.com/slvDev/esp32-ai
- research/tinystories/: https://github.com/slvDev/esp32-ai/tree/main/research/tinystories
- TinyStories dataset: https://huggingface.co/datasets/roneneldan/TinyStories
- 3Blue1Brown neural networks: https://www.youtube.com/playlist?list=PLZHQObOWTQDNU6R1_67000Dx_ZCJB-3pi